Vacuity and model-adequacy findings in a Lean formalization of conditional Navier–Stokes regularity

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Abstract

We report defects found in the Lean 4 companion formalization of two preprints on conditional regularity for the three-dimensional incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$ (10.5281/zenodo.24079623, 10.5281/zenodo.24080310). An attempt to prove the stated "analytic bridge" target led instead to a kernel-checked proof of its *negation*: the solution object did not encode the equations, so a zero velocity field with lifespan 1 counted as a solution. The same object shows that the cited Beale–Kato–Majda hypothesis was uninhabitable, making four conditional theorems vacuously true. Independently, the abstract data structure used for the "three-way equivalence" presupposed global regularity, so that equivalence carried no information. We describe each mechanism, the exact Lean theorems that establish it, and the corrected encoding: solutions that satisfy the Navier–Stokes equations on a maximal lifespan, the BKM and Kato–Ponce estimates as named cited hypotheses, and non-vacuity lemmas. **This note does not solve, and makes no claim about, the Clay Millennium Problem.** Global regularity for large data (Clay Statement (B)) remains open; in the corrected repository it is a single explicitly labelled open target.

1 What was claimed

The repository `BenFrohman/frohmanian-tether-conditional-regularity` accompanies the two preprints above. Its `README` and the file `AnalyticBridgeTarget.lean` stated:

1. an *open target* `analytic_bridge_unconditional`, described as "Clay Millennium Problem Statement (B) on the periodic 3-torus", with one intentional `sorry`, and explicitly flagged as using placeholder fields;

2. proved conditional results `clayB_iff_bkm_integrable`, `formB_implies_clayB`, `hypothesisB_implies_cla` and `formB_Hs_bound`, each taking a cited Beale–Kato–Majda (BKM) continuation hypothesis `BKMCriterion N` as an argument;

3. in `FrohmanianTripleArchitecture.lean` (the file archived with the Tether record), `hypothesisB_implies_` and `three_way_equivalence` over an abstract structure `NavierStokesUnifiedData`.

All proved declarations passed a kernel audit (`#print axioms` reporting only `propext`, `Classical.choice`, `Quot.sound`). That audit establishes that proofs are correct *relative to the definitions*; it says nothing about whether the definitions model Navier–Stokes, nor whether hypotheses can ever be supplied. The findings below are of exactly that second kind.

2 How the defects were caught

The user asked for a full proof of the open target. The procedure was:

1. Read the exact statement and every definition it depends on.
2. Ask which objects inhabit the hypothesis types. The structure `NSSolution` had fields `solves_NS : Prop` and `maximal : Prop`: these are *data* (arbitrary propositions chosen by whoever builds the object), not proof obligations. The lifespan `Tstar` was an arbitrary element of $[0, \infty]$.
3. Build the simplest object of the type with a finite lifespan, and prove the negation of the target in Lean, against the *literal* type of the declaration (read off with Lean's `type_of%` elaborator in the first commit, so that no hand restatement could drift from the original).
4. Re-use the same object against every hypothesis structure; check each abstract structure for a field that silently implies its conclusion.

Every finding is therefore a kernel-checked theorem, not an informal remark.

3 The counterexample and the refutation

Definition 1 (legacy zero object)**.** For any norm package N and viscosity ν, `zeroSolution N` is the legacy `NSSolution` with $u_0 = 0$, $u(t, x) = 0$ for all t, x, $T^\star = 1$, `solves_NS := True`, `maximal := True`. Every remaining field (initial condition, periodicity, C^ω smoothness of each $u(t)$, continuity of $t \mapsto N(u(t))$ for the three norm functionals) holds because every function involved is constant.

Theorem 2 (`analytic_bridge_target_statement_false`)**.** *The legacy target statement is false.*

Proof. Take N with all norms identically 0 (`zeroNorms`), $\nu = 1$, $s = 3 > 5/2$, $u_0 = 0$. Then u_0 is periodic and $N.\mathrm{Hs}(3, u_0) = 0 < \infty$, so all hypotheses hold. The target asserts $T^\star = \top$ for *every* solution object; for `zeroSolution` we have $T^\star = 1 \neq \top$. Contradiction. □

Theorem 3 (`not_nonempty_BKMCriterion`)**.** *For every N, the legacy type `BKMCriterion N` is empty.*

Proof. A term of `BKMCriterion N` maps any solution object whose function $t \mapsto N.\mathrm{gradLinf}(u(t))$ is interval-integrable on every $[0, T]$ to a proof of $T^\star = \top$. For `zeroSolution N 0` this function is the constant $N.\mathrm{gradLinf}(0)$, which is integrable on every bounded interval, but $T^\star = 1 \neq \top$. □

Remark 4 (consequence)**.** Each of `clayB_iff_bkm_integrable`, `formB_implies_clayB`, `hypothesisB_implies_` `formB_Hs_bound` has an argument of type `BKMCriterion N`. Since that type is empty, each theorem is of the form $\bot \to P$ and is vacuously true (`BKMCriterion_elim` derives `False` from any such argument). They were correct proofs of statements with no content.

4 The presupposed-regularity mechanism

`FrohmanianTripleArchitecture.NavierStokesUnifiedData` has a field

```
differential_bounds :  T > 0,  ( t  Icc 0 T, DifferentiableAt  vorticitySup t)
  ContinuousOn vorticitySup (Icc 0 T)  ContinuousOn strainSup (Icc 0 T)  ...
```

and defines Hypothesis B as: for every $T > 0$ there is C with $\int_0^t \|S\|_\infty \le C$ for all $t \in [0,T]$.

Theorem 5 (`hypothesisB_holds`, `clayStatementB_holds`)**.** *Every `NavierStokesUnifiedData` satisfies Hypothesis B and `clayStatementB`.*

Proof. Fix $T > 0$. By `differential_bounds`, f = strainSup is continuous on the compact interval $[0,T]$, hence $|f| \le K$ there for some K. For $t \in [0,T]$, $\int_0^t f \le \int_0^t \max(K,0) = \max(K,0)\, t \le \max(K,0)\, T$. This is Hypothesis B with $C = \max(K,0)T$. The existing theorem `hypothesisB_implies_clayState` then yields `clayStatementB`. □

Continuity of the strain on *every* $[0,T]$ is already a global-in-time regularity statement; it was placed in the structure as an input. Consequently the "three-way equivalence" between Hypothesis B, BKM integrability and Clay (B) is an equivalence between three propositions that are all provable for every object, and carries no information.

5 The fixes

5.1 A solution object that requires the equations

In `ClassicalNS.lean`, $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ is modelled by functions on $\mathbb{R}^3$ that are 1-periodic in each coordinate, and $\partial_i g(x) = Dg(x)e_i$ (Fréchet derivative). For ν, u_0, $u : \mathbb{R}\times\mathbb{R}^3 \to \mathbb{R}^3$, $p : \mathbb{R}\times\mathbb{R}^3 \to \mathbb{R}$ and $T^\star \in [0,\infty]$, the predicate `IsClassicalNS u0 u p Tstar` requires, with $L = [0,T^\star)$ and $L^\circ = (0,T^\star)$:

1. $u(0) = u_0$; $u(t)$ and $p(t)$ are 1-periodic and C^∞ for $t \in L$ (Mathlib's `ContDiff ℝ ⊤`, i.e. smooth, not analytic);
2. $(t,x) \mapsto u(t,x)$ and $(t,x) \mapsto p(t,x)$ are jointly C^∞ on $L^\circ \times \mathbb{R}^3$; u and $\nabla_x u$ are jointly continuous on $L \times \mathbb{R}^3$;
3. $\operatorname{div} u(t) = \sum_i \partial_i u_i(t) = 0$ for $t \in L$;
4. for $t \in L^\circ$, all x and j: $t \mapsto u_j(t,x)$ has derivative $\nu \sum_i \partial_i\partial_i u_j - \sum_i u_i \partial_i u_j - \partial_j p$ at t.

An `NSSolution u0` packages $\nu > 0$, $T^\star$, u, p, a proof of `IsClassicalNS`, and **maximality**: for every $T' > T^\star$ and every classical solution (v,q) on $[0,T')$ with the same data, v does not agree with u on L. The gradient norm is a definition, not a parameter: $\|\nabla f\|_\infty := \sup_{x\in[0,1]^3} \|Df(x)\|$ (operator norm; for periodic f the cell supremum is the global supremum).

Proposition 6 (non-vacuity; all proved in Lean)**.**

1. *`const_isClassicalNS, nonempty_NSSolution_zero`: for every $c \in \mathbb{R}^3$, $u \equiv c$, $p \equiv 0$ is a classical solution with $T^\star = \top$; in particular `NSSolution 0` is inhabited for $\nu > 0$.*
2. *`zero_field_is_global`: a solution with zero data that vanishes on its lifespan has $T^\star = \top$. Hence `old_counterexample_invalid`: there is no `NSSolution` with $u_0 = 0$, $u \equiv 0$ and $T^\star = 1$.*
3. *`linear_in_time_not_NS`: for any $T^\star > 0$, $u(t,x) = t\, e_0$, $p \equiv 0$ is* not *a classical solution. (At any $r \in (0,T^\star)$ the time derivative of the first component is 1, while every spatial derivative of the x-independent field vanishes, so the right-hand side is 0; uniqueness of derivatives gives $1 = 0$.)*

Proof of (2). If $T^\star \ne \top$ then $T^\star < \top$, and $(0,0)$ is a classical solution on $[0,\top)$ by (1) agreeing with u on L, contradicting maximality. □

5.2 Proved, not assumed: integrability for global solutions

Lemma 7 (`gradLinf_continuousOn`). *If $0 \le T$ and $T < T^\star$, then $t \mapsto \|\nabla u(t)\|_\infty$ is continuous on $[0, T]$.*

Proof. $G(t, x) = \|Du(t)(x)\|$ is continuous on $L \times \mathbb{R}^3$. Let π_T be the projection of $\mathbb{R}$ onto $[0, T]$; then $(t, x) \mapsto G(\pi_T t, x)$ is continuous on all of $\mathbb{R} \times \mathbb{R}^3$ because $[0, T] \subseteq L$. For a jointly continuous function and a compact set $K = [0, 1]^3$, $t \mapsto \sup_{x \in K}$ is continuous (Mathlib `IsCompact.continuous_sSup`). On $[0, T]$, $\pi_T t = t$. □

Theorem 8 (`global_implies_gradIntegral_finite`). *If $T^\star = \top$ then $\int_{[0,T)} \|\nabla u(t)\|_\infty \, dt < \infty$ for every T.*

Proof. For $T \le 0$ the domain is empty. For $T > 0$ the integrand is continuous on $[0, T]$ by the lemma, hence bounded by some C; the lower Lebesgue integral over $[0, T) \subseteq [0, T]$ is at most $C \cdot T < \infty$. □

5.3 BKM as a satisfiable cited hypothesis

Definition 9 (`BKMCriterion`, cited). For every maximal classical solution with $T^\star \neq \top$: $\int_{[0,T^\star)} \|\nabla u(t)\|_\infty \, dt = \infty$.

This is the Beale–Kato–Majda criterion in the form controlled by the full gradient (BKM 1984; Kato's local theory and uniqueness of classical solutions are used to pass between "cannot be continued" and "maximal in this class"). It is *not* proved in Lean. Unlike the legacy version it is not refuted by any object we can construct: its only quantified objects are genuine maximal solutions, and the legacy witness is excluded by `old_counterexample_invalid`. We cannot prove in Lean that it is consistent, since that would amount to formalizing BKM; we state this limitation explicitly.

Theorem 10 (conditional; proved in Lean).

1. *`gradIntegral_finite_implies_global`: BKM and finiteness of $\int_{[0,T)\cap L} \|\nabla u\|_\infty$ for all $T > 0$ imply $T^\star = \top$.*
2. *`clayB_iff_gradIntegral_finite`: under BKM, $T^\star = \top$ iff those integrals are finite for all $T > 0$.*
3. *`formB_implies_clayB`, `analyticBridge_of_formB`: Hypothesis B (a uniform bound $\int_{[0,T)\cap L} \|\nabla u\|_\infty \le C_\star(T)$ over all solutions) and BKM imply every maximal solution is global.*
4. *`formB_Hs_bound`: adding the cited Kato–Ponce inequality $\|u(t)\|_{H^s} \le \|u_0\|_{H^s} + \int_0^t C_s \|\nabla u\|_\infty \|u\|_{H^s}$ gives $\|u(t)\|_{H^s} \le \|u_0\|_{H^s} \exp(C_s \max(C_\star(T), 0))$ on $[0, T]$.*

Proof of (1). Suppose $T^\star \neq \top$ and put $T_0 = T^\star$ as a real number. BKM gives $\int_{[0,T_0)} = \infty$. Every $t \in [0, T_0)$ lies in $[0, T_0 + 1) \cap L$, so by monotonicity of the integral in the domain the hypothesis at $T = T_0 + 1$ is violated. □

Proof of (4). (3) gives $T^\star = \top$, so every t is in the lifespan. The integral Grönwall lemma (proved) applied with $\beta = C_s \|\nabla u\|_\infty$, continuous by the lemma above, gives $y(t) \le y(0) \exp \int_0^t \beta$. Since the integrand is continuous and nonnegative, $\int_0^t \|\nabla u\| = \int_{(0,t)} \|\nabla u\|$ and its ofReal equals the lower Lebesgue integral over $(0, t) \subseteq [0, T)$, which is $\le C_\star(T) \le \max(C_\star(T), 0)$; monotonicity of exp finishes. (The max is needed: a negative $C_\star(T)$ is only meaningful through ofReal, which clips it at 0.) □

The H^s norm is not defined in Mathlib on $\mathbb{T}^3$; it appears only as a documented parameter `HsR` inside the cited hypothesis `KatoPonceEnergy`. That structure is consistent (`kp_trivial`: HsR $\equiv 0$, $C_s = 0$ is an instance), but for that trivial instance the bound is trivially true; its content is for the true H^s norm.

5.4 Lifespan-corrected abstract data

In `FrohmanianTripleArchitectureLifespan.lean` the data carry $T^\star \in [0, \infty]$; every regularity requirement and the vorticity sign hold only for $T < T^\star$; the BKM field quantifies only over lifespan times; and `clayStatementB` is *defined* as $T^\star = \top$. `hypothesisB_implies_clayStatementB` (Grönwall on each $[0, t]$ with $t < T^\star$, then the BKM field) and `three_way_equivalence` are re-proved.

Proposition 11 (`blowupModel`; proved)**.** *$T^\star = 1$, $\|\omega\|_\infty(t) = \|S\|_\infty(t) = (1-t)^{-1}$ satisfies every field, and `clayStatementB` fails. Hence `clayStatementB_not_automatic`, `hypothesisB_not_automatic` and `old_trivialization_fails`.*

Proof. For $0 \le t \le T < 1$: $(1-t)^{-1} \ge 0$; it is differentiable with derivative $(1-t)^{-2} = S \cdot M$, so $M' \le SM$ holds with equality; both functions are continuous on $[0, T]$. For the BKM field we must show its premise fails. Given any C, put $a = e^{-(|C|+1)} \in (0, 1)$ and $t = 1 - a \in [0, 1)$. By the substitution $x = 1 - \tau$, $\int_0^t (1-\tau)^{-1} d\tau = \int_a^1 x^{-1} dx = \log(1/a) = |C| + 1 > C$. So no C bounds the vorticity integrals on $[0, 1)$, the premise is false and the implication holds. Finally $1 \neq \top$. If Hypothesis B held for this model, the re-proved theorem would give $T^\star = \top$; so it fails. □

These are abstract scalar models (as was the original structure), not Navier–Stokes solutions; they show only that the implications are no longer contentless.

5.5 Legacy material

Nothing was deleted. `FrohmanianTripleArchitecture.lean` is byte-identical. In `AnalyticBridgeTarget.lean` the header now marks the file LEGACY, and the former theorem-with-placeholder (a false statement) is kept verbatim as the proposition `analytic_bridge_unconditional_statement`, which the refutation file negates.

6 What remains open

The honest target, `ClassicalNS.analytic_bridge_unconditional : AnalyticBridgeStatement`, says: for every $\nu > 0$ and every smooth, periodic, divergence-free u_0, every maximal classical solution is global with $\int_0^T \|\nabla u\|_\infty < \infty$ for all T. Combined with local existence and uniqueness of classical solutions (standard, not formalized), this is Clay Statement (B) for the periodic problem. Its single `sorry` is labelled OPEN, it is excluded from the forbidden-token scan and from every axiom audit, and `#print axioms` on it reports `sorryAx`. The mathematical gap is unchanged: no known argument controls the critical quantity $\int_0^T \|\nabla u\|_\infty$ (or $\int_0^T \|\omega\|_\infty$) for large data from the energy-class bounds $L^\infty_t L^2_x \cap L^2_t \dot{H}^1_x$. **Nothing in this note or the repository proves global regularity, blow-up, or any new fact about Navier–Stokes.**

Other limitations: the solution class is classical (pointwise equations, joint smoothness on the open lifespan); the BKM and Kato–Ponce statements are cited, not proved; Hypothesis B is of Clay strength and its truth is open, so theorems conditional on it are conditional on an open assumption.

7 Reproducibility

- Repository: `https://github.com/BenFrohman/frohmanian-tether-conditional-regularity`, branch `analytic-bridge-refutation`; Lean sources at commit `cb927bd302b144982bc2b277423e1aeb9a8b7` The first refutation (against the literal `type_of%` of the legacy declaration) is commit `1a89946`.
- Toolchain `leanprover/lean4:v4.30.0-rc1`; Mathlib `v4.30.0-rc1` (pinned in `lean-toolchain`, `lakefile.toml`, `lake-manifest.json`).
- One-command rebuild and audit:

```
lake exe cache get && lake build && ./scripts/kernel_check.sh
```

- Audit result: 52 declarations (14 in `scripts/PrintAxioms.lean`, 38 in `scripts/PrintAxiomsBridge.lean`), each reported as

```
depends on axioms: [propext, Classical.choice, Quot.sound]
```

  The only build warning is the intentional placeholder in `ClassicalNSOpenTarget.lean`.

References

- J. T. Beale, T. Kato, A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, Comm. Math. Phys. 94 (1984), 61–66.
- T. Kato, G. Ponce, Commutator estimates and the Euler and Navier–Stokes equations, Comm. Pure Appl. Math. 41 (1988), 891–907.
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